Manoj Kumar Thapa

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UK Graduate Route visa valid to 2028 – no sponsorship required

EDUCATION

MSc Artificial Intelligence, Aston University, UK – Distinction (70.37%) Jan 2025 – Jul 2026
BE Computer Science, Dr. Ambedkar Institute of Technology, India – CGPA 9.45/10 Aug 2017 – Sep 2021

PUBLICATION

Vinutha H., **M. K. Thapa**, P. Pandey, N. Pun, N. Paudel. *Indian Currency Recognition for Visually Impaired using Deep Learning Technique*. IJIRCCE, 9(7), July 2021. DOI: 10.15680/IJIRCCE.2021.0907159 · PDF

PROJECTS

FruitGuard AI – MSc Dissertation | *PyTorch, YOLOv8, ONNX Runtime* | Demo Aug 2025 – Apr 2026

- Fine-tuned a YOLOv8 model with a two-phase transfer-learning pipeline to **96.3%** classification accuracy (52 of 54) from only **207** labelled images, a **34x** reduction against the **7,108** images used for pre-training.
- Beat the COCO-pretrained baseline by **+6.03 pp** mAP@0.5:0.95 (0.5854 to 0.6457) and **+3.12 pp** precision.
- Optimized inference with ONNX Runtime to real-time **73 FPS**, deployed as a live Hugging Face demo.

Planck AI – Agentic Research Assistant | *Python, FastAPI, React, Docker* | GitHub · Demo Sep 2025 – Nov 2025

- Built an LLM agent in Python without an agent framework, orchestrating 4 tools (web search, code execution in 7 languages, PDF/URL reading, image analysis) across up to 8 planning steps.
- Designed mid-run failover across Groq, Gemini and NVIDIA behind one OpenAI-compatible interface, sustaining **30/30** task completion under fault injection with the primary provider disabled on every call.
- Built an evaluation harness with a 30-question golden set and a rubric committed to Git before the first run, measuring **80%** correct tool selection, **2.7 s** median latency and no incorrect answer among the **20** tasks the rubric could grade.

CogniGraph – Graph RAG Explorer | *Python, ChromaDB, NetworkX, OCR* | GitHub · Demo Jun 2025 – Jul 2025

- Built a knowledge-graph RAG app that turns documents, including scanned PDFs via OCR, into an interactive 3D graph and answers only from passages retrieved from a ChromaDB vector database.
- Rebuilt the evaluation set after finding self-written test questions overstated accuracy by **35 points** (90% to 55% hit@1), then benchmarked the corrected set at **95% hit@3** against **80%** for a keyword baseline on the same questions.
- Proved the embedding model silently discards text past ∼**1,400 characters**, showed the sweep favouring 2,000-character chunks was confounded by corpus size, and set chunk size to **1,000**, the largest that fits the window.

NeuroArc – AI Job Application Assistant | *FastAPI, React, Groq API, OCR* | GitHub · Demo Nov 2025 – Dec 2025

- Built an end-to-end assistant that parses CVs (PDF/DOCX, OCR fallback), pulls live UK jobs from the Reed API, scores CV–job fit and exports a tailored CV as PDF.
- Stopped the model inventing its own evidence: it returns requirement–evidence pairs as structured JSON, and code discards any requirement whose quoted evidence is not in the CV before computing the weighted score.

EXPERIENCE

Mentor, MSc Artificial Intelligence, Aston University | *Birmingham, UK* Aug 2025 – Jan 2026

- Mentored **5** MSc AI students in Python and machine learning, one-to-one, every week.
- Focused sessions on the three blockers that recurred most: setting up Python environments, debugging training loops that would not converge, and choosing a simple baseline before a complex model.

Software Engineer, Accenture | *Bengaluru, India* Oct 2021 – May 2022

- Cut p95 REST API latency from roughly **850 ms to 510 ms** across **18 endpoints**, a **40%** reduction, by eliminating **6** N+1 query sites and rewriting the SQL behind them in Java Spring Boot.
- Owned the business logic and the PostgreSQL/MySQL data-access layer for that service, shipping production REST APIs in a **7-engineer** delivery team on 2-week sprints.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, TypeScript
LLMs & Agents: Tool/function calling, agent loops, LangGraph, RAG, prompt engineering, structured JSON output, multi-provider LLM fallback (Groq, Gemini, NVIDIA; OpenAI-compatible APIs)
Retrieval & Data: ChromaDB (vector database), embeddings, semantic search, OCR, PostgreSQL, MySQL
ML & Computer Vision: PyTorch, YOLOv8, transfer learning, fine-tuning, ONNX Runtime, quantization
Evaluation: Golden-set design, fault injection, baseline controls, latency benchmarking
Backend & Frontend: FastAPI, Pydantic, Spring Boot, REST API design, React, Tailwind CSS
Tools & Deployment: Docker, Docker Compose, Git, Hugging Face Spaces, Render

CERTIFICATIONS

AI Agents Intensive (Google, Kaggle) | Machine Learning Specialization (Stanford) | Deep Learning Specialization (DeepLearning.AI)